Day 68 - Final Optimization & Bug Fixing

Bug Fix Report and Final System Validation

Objective and result

The codebase was stabilized and validated for the synthetic hiring-pipeline demonstration. The full automated suite passed after correcting a test-isolation issue and directing temporary test files to an accessible workspace folder.

Bug fix

- **Issue:** `tests/test_section_builder.py` ran as a top-level script during pytest collection and overwrote the tracked `data/labeled_resumes/sample.json` fixture.
- **Change:** Converted it into a normal pytest round-trip test using `tmp_path`. It now checks save/load behavior without changing repository data.
- **Result:** The original sample fixture remained unchanged after the final full-suite run. No scoring or hiring-decision logic was modified.

Validation results

- **Automated tests:** 679 passed; 8 subtests passed; 0 failures in the final run.
- **Python compilation:** Passed for 8 source/test folders and 18 root-level `main*.py` entry points.
- **Diff hygiene:** `git diff --check` reported no whitespace errors.
- **Windows test setup:** The first run hit access-denied errors because the sandbox TEMP folder was restricted. Setting `TEMP` and `TMP` to a workspace-owned folder resolved the environment issue; the final suite completed successfully.

End-to-end output consistency

- Fresh run covered all 4 candidate profiles referenced by the synthetic dataset manifest and wrote its report to a temporary validation path.
- All expected stages were present, from structured resume intake through final recommendation. Stage scores were finite and within the 0-100 range.
- Each unified score matched the sum of its weighted round contributions. The run produced 3 “Selected” and 1 “Hold / Review” demo outcomes.
- The report clearly labels resume intake as structured fixtures and machine-test evaluation as fixture-scored; it does not claim a production upload or code-execution path.

Performance and Release Readiness

Measured on 05 November 2026 in the local Windows workspace

Performance measurements

- **Day 63 full synthetic pipeline:** 4 profiles completed in 1.0223 seconds, excluding model construction and startup.
- **Warm scoring benchmark:** 8 resume files; first warm-up/first score 1.0767 seconds; total warm scoring 1.0215 seconds; average 0.1277 seconds per resume.
- **Embedding cache:** 115 hits, 51 misses, 51 stored entries during the benchmark.
- **Tuning decision:** Kept the existing embedding-cache design. The measured local run did not show a need for a risky late-stage scoring change; the full synthetic run remained near one second.

These are single local demonstration measurements, not a load test or a production latency guarantee. Results can vary with hardware, model-cache state, and input length.

Release status

Validated for a local synthetic demonstration. The test suite, Python compilation, end-to-end data flow, score bounds, and output arithmetic passed. The result should be treated as a demo release candidate rather than a production hiring system.

Known scope limits

- The Day 63 runner consumes structured synthetic resume JSON; its run does not exercise a production upload endpoint or parser integration.
- HR and technical stages use deterministic demonstration heuristics and supplied response/rubric fixtures. The machine-test stage scores fixture inputs and does not execute candidate code.
- All profiles are fictional. Human review remains necessary before any real employment decision; this validation does not establish legal, fairness, security, or production-scale readiness.

Files changed

Only `tests/test_section_builder.py` was changed. The Day 63 validation report and pytest temporary files were kept under `.tmp` during verification; the existing Day 63 report was not overwritten.

Validation conclusion: the current codebase passes its full automated suite and the synthetic end-to-end checks after isolating the test fixture side effect.